

openapi: 3.1.0

info:

title: Cloud Health Office – Prior Authorization API

version: 4.0.0

description: |

FHIR R4 Prior Authorization API with CDS Hooks for CMS-0057-F compliance.

Implements electronic prior authorization aligned with the Da Vinci Prior Authorization Support (PAS) Implementation Guide and Coverage Requirements Discovery (CRD) / Documentation Templates and Rules (DTR) for PARDD.

****CMS-0057-F requires**** payers to support electronic prior auth submission, status inquiry, and real-time coverage requirements discovery by January 1, 2

****Key Capabilities:****

- FHIR Claim-based prior auth submission (replaces X12 278)
- Real-time auth status inquiry via ClaimResponse
- CDS Hooks for coverage requirements discovery (CRD)
- Bidirectional X12 278 ↔ FHIR mapping
- SLA tracking (urgent: 24h, standard: 7 days per CMS timelines)
- Clinical attachment support (DocumentReference, Binary)
- Clearinghouse packaging for hybrid X12/FHIR workflows
- Decision timeline audit trail

****References:****

- CMS-0057-F Final Rule (Prior Authorization Requirements)
- Da Vinci PAS (Prior Authorization Support) IG
- Da Vinci CRD (Coverage Requirements Discovery) IG
- Da Vinci DTR (Documentation Templates and Rules) IG
- CDS Hooks Specification v2.0
- X12 278 Health Care Services Review

contact:

name: Aurelianware, Inc.

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license:

name: Apache 2.0

url: <https://www.apache.org/licenses/LICENSE-2.0>

servers:

- url: <https://api.cloudhealthoffice.com/fhir/r4>
description: Production
- url: <https://sandbox.cloudhealthoffice.com/fhir/r4>
description: Sandbox
- url: <http://localhost:3000/fhir/r4>

```

    description: Local development

security:
  - smart_on_fhir: [user/Claim.write, user/ClaimResponse.read]

tags:
  - name: Prior Authorization
    description: Submit and check prior authorization requests
  - name: CDS Hooks
    description: Real-time coverage requirements discovery (CRD/DTR)
  - name: Attachments
    description: Clinical documentation attachments
  - name: X12 Mapping
    description: Bidirectional X12 278 ↔ FHIR conversion
  - name: SLA
    description: Authorization decision SLA tracking

paths:

# — Submit Prior Auth —————

/Claim/$submit:
  post:
    operationId: submitPriorAuth
    summary: Submit a prior authorization request
    description: |
      Submit a FHIR Claim resource with `use: preauthorization` to request
      prior authorization. Replaces the X12 278 request workflow.

      The payer processes the request and returns a ClaimResponse with the
      authorization decision (approved, denied, pending for review).

      If the request is pending, the response includes an `authorizationId`
      for status polling and any requests for additional information (RFAI).
    tags: [Prior Authorization]
    requestBody:
      required: true
      content:
        application/fhir+json:
          schema:
            $ref: '#/components/schemas/PriorAuthRequest'
          example:
            resourceType: Claim
            status: active
            type:
              coding:
                - system: "http://terminology.hl7.org/CodeSystem/claim-type"

```

```

        code: institutional
    use: preauthorization
    patient:
        reference: "Patient/pat-12345"
    created: "2026-02-16"
    provider:
        reference: "Practitioner/prov-npi-1234567890"
    priority:
        coding:
            - system: "http://terminology.hl7.org/CodeSystem/processpriorit
              code: normal
    insurance:
        - sequence: 1
          focal: true
          coverage:
              reference: "Coverage/cov-001"
    diagnosis:
        - sequence: 1
          diagnosisCodeableConcept:
              coding:
                  - system: "http://hl7.org/fhir/sid/icd-10-cm"
                    code: "M17.11"
                    display: "Primary osteoarthritis, right knee"
    item:
        - sequence: 1
          productOrService:
              coding:
                  - system: "http://www.ama-assn.org/go/cpt"
                    code: "27447"
                    display: "Total knee replacement"
          servicedDate: "2026-03-15"
          locationCodeableConcept:
              coding:
                  - system: "https://www.cms.gov/Medicare/Coding/place-of-ser
                    code: "21"
                    display: "Inpatient Hospital"
          quantity:
              value: 1
    supportingInfo:
        - sequence: 1
          category:
              coding:
                  - code: "info"
          valueString: "Patient has failed conservative treatment includi
responses:
    '200':
        description: Authorization decision

```

```

content:
  application/fhir+json:
    schema:
      $ref: '#/components/schemas/PriorAuthResponse'
    examples:
      approved:
        summary: Approved
        value:
          resourceType: ClaimResponse
          id: "auth-resp-001"
          status: active
          type:
            coding: [{ system: "http://terminology.hl7.org/CodeSystem/c
            use: preauthorization
          patient:
            reference: "Patient/pat-12345"
            created: "2026-02-16T10:30:00Z"
            insurer:
              reference: "Organization/payer-001"
            outcome: complete
            disposition: "Authorized"
            preAuthRef: "AUTH-2026-00456"
            preAuthPeriod:
              start: "2026-03-01"
              end: "2026-06-01"
            item:
              - itemSequence: 1
                adjudication:
                  - category:
                      coding: [{ code: "submitted" }]
                    reason:
                      coding: [{ code: "approved", display: "Approved" }]
      pending:
        summary: Pending for review
        value:
          resourceType: ClaimResponse
          id: "auth-resp-002"
          status: active
          use: preauthorization
          outcome: queued
          disposition: "Pending – additional clinical documentation requ
          preAuthRef: "AUTH-2026-00457"
          extension:
            - url: "https://cloudhealthoffice.com/fhir/ext/rfai"
              valueString: "Please provide operative report and PT note
'400':
  description: Invalid request

```

```
    content:
      application/fhir+json:
        schema:
          $ref: '#/components/schemas/OperationOutcome'
```

— Check Auth Status —————

/ClaimResponse/{authorizationId}:

```
get:
  operationId: getAuthorizationStatus
  summary: Check prior authorization status
  description: |
    Poll for the status of a pending authorization request.
    Returns the current ClaimResponse with decision status,
    any RFAI (request for additional information), and SLA timeline.
  tags: [Prior Authorization]
  parameters:
    - name: authorizationId
      in: path
      required: true
      schema:
        type: string
      example: "AUTH-2026-00457"
  responses:
    '200':
      description: Current authorization status
      content:
        application/fhir+json:
          schema:
            $ref: '#/components/schemas/PriorAuthResponse'
    '404':
      $ref: '#/components/responses/NotFound'
```

— Search Authorizations —————

/ClaimResponse:

```
get:
  operationId: searchAuthorizations
  summary: Search prior authorizations
  description: |
    Search for prior authorizations by patient, provider, status, or date.
    CMS-0057-F requires payers to make authorization status available
    to both providers and patients.
  tags: [Prior Authorization]
  parameters:
    - name: patient
      in: query
```

```

    schema:
      type: string
      example: "Patient/pat-12345"
  - name: provider
    in: query
    schema:
      type: string
  - name: outcome
    in: query
    schema:
      type: string
      enum: [complete, queued, partial, error]
  - name: created
    in: query
    schema:
      type: string
      description: Date range filter
  - name: _count
    in: query
    schema:
      type: integer
      default: 20
responses:
  '200':
    description: Bundle of ClaimResponse resources
    content:
      application/fhir+json:
        schema:
          $ref: '#/components/schemas/Bundle'

```

— CDS Hooks – Coverage Requirements Discovery —————

```

/cds-services:
  get:
    operationId: getCDSServices
    summary: CDS Hooks service discovery
    description: |
      Returns the list of available CDS Hooks services.
      Used by EHR systems to discover what decision support
      is available during the ordering workflow.
    tags: [CDS Hooks]
    security: []
    responses:
      '200':
        description: CDS service catalog
        content:
          application/json:

```

```

    schema:
      type: object
      properties:
        services:
          type: array
          items:
            type: object
    example:
      services:
        - id: "prior-auth-crd"
          hook: "order-sign"
          title: "Coverage Requirements Discovery"
          description: "Check if prior authorization is required for th
          prefetch:
            patient: "Patient/{{context.patientId}}"
            coverage: "Coverage?patient={{context.patientId}}&status=ac
        - id: "prior-auth-check"
          hook: "appointment-book"
          title: "Prior Auth Pre-Check"
          description: "Verify prior authorization before scheduling"

```

/cds-services/prior-auth-crd:

post:

```

  operationId: coverageRequirementsDiscovery
  summary: Check if prior auth is required (CRD)
  description: |
    CDS Hooks endpoint for Coverage Requirements Discovery.
    Called by the EHR during the `order-sign` hook to determine
    if the ordered service requires prior authorization.

```

Returns CDS Cards with:

- Whether prior auth is required
- What documentation is needed (DTR link)
- Estimated approval timeline
- Alternative services that don't require auth

tags: [CDS Hooks]

requestBody:

required: true

content:

application/json:

schema:

\$ref: '#/components/schemas/CDSHookRequest'

example:

hook: "order-sign"

hookInstance: "d1577c69-dfbe-44ad-ba6d-3e05e953b2ea"

context:

userId: "Practitioner/prov-npi-1234567890"

```

    patientId: "pat-12345"
    draftOrders:
      resourceType: Bundle
      entry:
        - resource:
            resourceType: ServiceRequest
            code:
              coding:
                - system: "http://www.ama-assn.org/go/cpt"
                  code: "27447"
                  display: "Total knee replacement"
            subject:
              reference: "Patient/pat-12345"
    prefetch:
      patient:
        resourceType: Patient
        id: "pat-12345"
      coverage:
        resourceType: Bundle
        entry:
          - resource:
              resourceType: Coverage
              id: "cov-001"
              subscriberId: "MBR-2024-001"
  responses:
    '200':
      description: CDS Hook response with cards
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/CDSHookResponse'
          example:
            cards:
              - uuid: "card-pa-001"
                summary: "Prior authorization required for CPT 27447"
                detail: "Total knee replacement requires prior authorization"
                indicator: warning
                source:
                  label: "Cloud Health Office"
                  url: "https://cloudhealthoffice.com"
                suggestions:
                  - label: "Submit prior authorization"
                    actions:
                      - type: create
                        description: "Create prior auth request"
                        resource:
                          resourceType: Task

```



```
        intent: proposal
        code:
          coding:
            - code: "prior-auth"
links:
  - label: "Complete DTR questionnaire"
    url: "https://portal.cloudhealthoffice.com/dtr/questionnaire"
    type: smart
  - label: "View coverage policy"
    url: "https://portal.cloudhealthoffice.com/policies/joint"
    type: absolute
```

— Attachments

/DocumentReference:

```
post:
  operationId: submitAttachment
  summary: Submit clinical documentation for prior auth
  description: |
    Submit supporting clinical documentation (operative reports,
    PT notes, imaging, lab results) as a FHIR DocumentReference
    linked to an existing prior authorization request.
  tags: [Attachments]
  requestBody:
    required: true
    content:
      application/fhir+json:
        schema:
          type: object
        example:
          resourceType: DocumentReference
          status: current
          type:
            coding:
              - system: "http://loinc.org"
                code: "11506-3"
                display: "Progress note"
          subject:
            reference: "Patient/pat-12345"
          context:
            related:
              - reference: "ClaimResponse/AUTH-2026-00457"
          content:
            - attachment:
                contentType: "application/pdf"
                url: "Binary/att-001"
                title: "Physical Therapy Progress Notes"
```

```
        creation: "2026-02-10"
responses:
  '201':
    description: Attachment stored and linked to authorization
```

— X12 278 Mapping —————

```
/x12/278/to-fhir:
  post:
    operationId: mapX12ToFhir
    summary: Convert X12 278 to FHIR prior auth
    description: |
      Converts an inbound X12 278 Health Care Services Review
      request to a FHIR Claim resource. Used for clearinghouse
      integration and hybrid X12/FHIR workflows.
    tags: [X12 Mapping]
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/X12_278_Request'
          example:
            transactionSetId: "278"
            controlNumber: "123456789"
            submitterName: "ABC Medical Group"
            submitterIdentifier: "1234567890"
            receiverName: "Cloud Health Office"
            receiverIdentifier: "9876543210"
            subscriberId: "MBR-2024-001"
            patientLastName: "Smith"
            patientFirstName: "Jane"
            patientDob: "1985-03-15"
            serviceType: "surgical"
            procedureCode: "27447"
            procedureCodeQualifier: "HC"
            diagnosisCode: "M17.11"
            facilityCode: "21"
            requestDate: "2026-02-16"
            serviceFromDate: "2026-03-15"
            serviceToDate: "2026-03-15"
            quantity: 1
            certificationAction: "AR"
    responses:
      '200':
        description: Mapped FHIR Claim resource
        content:
```

```

        application/fhir+json:
          schema:
            $ref: '#/components/schemas/PriorAuthRequest'

/x12/278/from-fhir:
  post:
    operationId: mapFhirToX12
    summary: Convert FHIR prior auth response to X12 278
    description: |
      Converts a FHIR ClaimResponse back to X12 278 format
      for clearinghouse transmission. Used for outbound responses
      to providers who submit via X12.
    tags: [X12 Mapping]
    requestBody:
      required: true
      content:
        application/fhir+json:
          schema:
            $ref: '#/components/schemas/PriorAuthResponse'
    responses:
      '200':
        description: X12 278 response
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/X12_278_Response'

```

— SLA Tracking —————

```

/prior-auth-sla/{authorizationId}:
  get:
    operationId: getSlaStatus
    summary: Get authorization decision SLA status
    description: |
      Returns the SLA timeline for an authorization request.
      CMS-0057-F mandates: urgent requests within 24 hours,
      standard requests within 7 calendar days.
    tags: [SLA]
    parameters:
      - name: authorizationId
        in: path
        required: true
        schema:
          type: string
    responses:
      '200':
        description: SLA status

```

```
content:
  application/json:
    schema:
      $ref: '#/components/schemas/PriorAuthSLA'
    example:
      authorizationId: "AUTH-2026-00457"
      priority: "standard"
      submittedAt: "2026-02-16T10:00:00Z"
      slaDeadline: "2026-02-23T10:00:00Z"
      hoursRemaining: 168
      hoursElapsed: 0.5
      isCompliant: true
      status: "pending_review"
```

```
components:
  schemas:
    PriorAuthRequest:
      type: object
      description: FHIR Claim with use=preauthorization
      properties:
        resourceType:
          type: string
          const: Claim
        status:
          type: string
        type:
          type: object
        use:
          type: string
          const: preauthorization
        patient:
          type: object
        created:
          type: string
        provider:
          type: object
        priority:
          type: object
        insurance:
          type: array
        diagnosis:
          type: array
        item:
          type: array
        supportingInfo:
          type: array
```

PriorAuthResponse:

```
type: object
description: FHIR ClaimResponse with authorization decision
properties:
  resourceType:
    type: string
    const: ClaimResponse
  id:
    type: string
  status:
    type: string
  use:
    type: string
    const: preauthorization
  patient:
    type: object
  created:
    type: string
  insurer:
    type: object
  outcome:
    type: string
    enum: [complete, queued, partial, error]
  disposition:
    type: string
  preAuthRef:
    type: string
    description: Authorization number
  preAuthPeriod:
    type: object
    properties:
      start:
        type: string
      end:
        type: string
```

CDSHookRequest:

```
type: object
properties:
  hook:
    type: string
    enum: [order-sign, order-select, appointment-book]
  hookInstance:
    type: string
    format: uuid
  context:
    type: object
```

```
properties:
  userId:
    type: string
  patientId:
    type: string
  draftOrders:
    type: object
  appointments:
    type: object
prefetch:
  type: object
```

```
CDSHookResponse:
  type: object
  properties:
    cards:
      type: array
    items:
      type: object
      properties:
        uuid:
          type: string
        summary:
          type: string
        detail:
          type: string
        indicator:
          type: string
          enum: [info, warning, critical]
        source:
          type: object
        suggestions:
          type: array
        links:
          type: array
```

```
X12_278_Request:
  type: object
  properties:
    transactionSetId:
      type: string
    controlNumber:
      type: string
    submitterName:
      type: string
    submitterIdentifier:
      type: string
```

```
subscriberId:
  type: string
patientLastName:
  type: string
patientFirstName:
  type: string
patientDob:
  type: string
procedureCode:
  type: string
diagnosisCode:
  type: string
facilityCode:
  type: string
requestDate:
  type: string
serviceFromDate:
  type: string
certificationAction:
  type: string
```

X12_278_Response:

```
type: object
properties:
  transactionSetId:
    type: string
  controlNumber:
    type: string
  certificationAction:
    type: string
    enum: [A1, A2, A3, A4, A6, CT]
    description: "A1=Approved, A2=Approved with mods, A3=Denied, A4=Pended,
  authorizationNumber:
    type: string
  effectiveDateRange:
    type: object
  reviewDecisionReason:
    type: string
```

PriorAuthSLA:

```
type: object
properties:
  authorizationId:
    type: string
  priority:
    type: string
    enum: [urgent, standard]
```

```
submittedAt:
  type: string
  format: date-time
slaDeadline:
  type: string
  format: date-time
decidedAt:
  type: string
  format: date-time
hoursRemaining:
  type: number
hoursElapsed:
  type: number
isCompliant:
  type: boolean
status:
  type: string
  enum: [pending_review, approved, denied, pended_rfai, expired]
```

```
Bundle:
  type: object
  properties:
    resourceType:
      type: string
      const: Bundle
    type:
      type: string
    total:
      type: integer
    entry:
      type: array
```

```
OperationOutcome:
  type: object
  properties:
    resourceType:
      type: string
      const: OperationOutcome
    issue:
      type: array
```

```
responses:
  NotFound:
    description: Authorization not found
    content:
      application/fhir+json:
        schema:
```


\$ref: '#/components/schemas/OperationOutcome'

securitySchemes:

smart_on_fhir:

type: oauth2

description: SMART on FHIR for provider EHR integration

flows:

authorizationCode:

authorizationUrl: https://login.cloudhealthoffice.com/oauth2/authorize

tokenUrl: https://login.cloudhealthoffice.com/oauth2/token

scopes:

user/Claim.write: Submit prior authorization requests

user/Claim.read: Read authorization requests

user/ClaimResponse.read: Read authorization decisions

user/DocumentReference.write: Submit clinical attachments

openid: OpenID Connect

fhirUser: FHIR user identity